

Lab – 2

AIM : Prepare SRS document for project and Write functional requirements for the systems given in Case Study document and Submit along with SRS of project.

Software Requirements Specification

for

EquiTrack

Version 1.0

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Full Team	Initial Draft for EquiTrack	17/07/26

1 Introduction

1.1 Purpose

The purpose of this **Software Requirements Specification (SRS)** is to define the functional and non-functional requirements for the EquiTrack Stock Market Analysis and Tracking System. This document serves as a reference for developers, testers, stakeholders, and end users during development and maintenance.

1.2 Scope

EquiTrack is a web-based stock market information application that enables users to monitor market trends and company information. Key capabilities: user registration and login, stock search, personalized watchlist, stock charts, company details, suggested stocks, top gainers/losers, and a dashboard. The system is informational and educational only; it does **not** support trading or financial transactions.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface
JWT	JSON Web Token
MERN	MongoDB; Express.js; React.js; Node.js
Watchlist	A personalized list of stocks saved by the user
Dashboard	Home page displaying market information

2 Overall Description

2.1 Product Perspective

EquiTrack is a single-page web application built on the **MERN stack**. The frontend (React) communicates with the backend (Node/Express) via REST APIs. The backend fetches market data from external stock APIs and stores user data in MongoDB.

2.2 Product Functions

Major system functions:

- **User Registration**
- **User Login**
- **User Profile Management**
- **Dashboard**
- **Stock Search**
- **Watchlist Management**
- **Stock Charts**
- **Suggested Stocks**
- **Top Gainers and Top Losers**
- **Company Details**
- **Logout**

2.3 User Characteristics

Intended users: Individuals interested in monitoring stock market information.

Registered user capabilities:

- Create an account
- Log in
- Search stocks
- View stock charts

- Manage a personal watchlist
- View company details
- View suggested stocks
- View top gainers and top losers
- Update profile information
- Logout securely

3 Funtional Requirements

R.1 User Management

Description: Manage user accounts, authentication, and profiles.

R.1.1 User Registration

- **Input:** Full Name; Email Address; Password
- **Output:** Registration successful message
- **Processing:** Validate inputs; check email uniqueness; store user securely

R.1.2 User Login

- **Input:** Email Address; Password
- **Output:** Redirect to Dashboard
- **Processing:** Validate credentials; authenticate user; create authenticated session (JWT)

R.1.3 View Profile

- **Input:** Click Profile
- **Output:** User profile information
- **Processing:** Retrieve and display user details from database

R.1.4 Update Profile

- **Input:** Name; Password (optional)
- **Output:** Profile updated successfully message
- **Processing:** Validate updates; save changes; display updated profile

R.1.5 Logout

- **Input:** Click Logout
- **Output:** Redirect to Login page
- **Processing:** End user session; clear authentication; redirect to login

R.2 Dashboard

Description: Provide market overview and quick access to features after login.

R.2.1 View Dashboard

- **Input:** User logs in
- **Output:** Market overview; suggested stocks; top gainers; top losers
- **Processing:** Retrieve latest market data; display dashboard widgets

R.2.2 View Top Gainers and Top Losers

- **Input:** Click Top Gainers or Top Losers
- **Output:** List of top gaining stocks; list of top losing stocks
- **Processing:** Retrieve latest market data; display lists

R.2.3 View Suggested Stocks

- **Input:** Open Dashboard
- **Output:** Suggested stock list
- **Processing:** Retrieve recommendation data; display suggestions

R.2.4 View Suggested Stock Details

- **Input:** Select a suggested stock
- **Output:** Company details
- **Processing:** Retrieve and display company information

R.3 Stock Management

Description: Search, analyze, and manage stock information.

R.3.1 Search Stock

- **Input:** Company name; Stock symbol
- **Output:** Matching stock list
- **Processing:** Read search keyword; query API/database; display matches

R.3.2 Manage Watchlist

- **Input:** Add to Watchlist; Remove from Watchlist
- **Output:** Stock added/removed success message
- **Processing:** Verify stock; check duplicates; add/remove; update watchlist

R.3.3 View Watchlist

- **Input:** Click Watchlist
- **Output:** List of saved stocks
- **Processing:** Retrieve watchlist from database; display saved stocks

R.3.4 View Stock Chart

- **Input:** Select a stock
- **Output:** Stock price chart
- **Processing:** Retrieve historical data; render chart

R.3.5 Change Chart Duration

- **Input:** Select time period (5 minutes; 1 day; 1 week; 1 month)
- **Output:** Updated stock chart
- **Processing:** Retrieve data for selected duration; refresh chart

R.3.6 View Company Information

- **Input:** Select a company
- **Output:** Company name; stock symbol; current price; day high; day low; description; price chart
- **Processing:** Retrieve company info; latest prices; historical data; display details

4 Other Non-functional Requirements

4.1 Performance

- **Dashboard** shall load within **5 seconds** under normal network conditions.
- **Search results** shall display within **2 seconds**.
- **Stock charts** shall load within **5 seconds**.

4.2 Security

- **Passwords** shall be encrypted before storage (use bcrypt).
- **JWT** shall be used for authentication.
- **HTTPS** shall be enforced for all communication.

4.3 Reliability

- The application shall maintain **data consistency**.
- User sessions remain active until logout or session timeout.

4.4 Usability

- The application shall provide a **responsive** and **user-friendly** interface.
- Navigation shall be **simple** and **intuitive**.

4.5 Maintainability

- The application shall follow a **modular architecture**.
- Source code shall be **documented** and follow coding standards.

5 Design Constraints

5.1 Design Constraints

- Active internet connection required.
- Market data depends on third-party stock market APIs.
- Users cannot perform actual stock trading.
- Each email address must be unique.
- Personalized features require user authentication.

5.2 Assumptions and Dependencies

- Users have access to a modern web browser.
- Internet connectivity is available.
- External stock market APIs are operational.
- MongoDB database services are available.
- Backend server is continuously accessible.

6 Assumptions and Dependencies

Component	Technology / Tool
Frontend	React.js; HTML; CSS; JavaScript
Backend	Node.js; Express.js
Database	MongoDB
Authentication	JWT; bcrypt
Development Tools	Visual Studio Code; Git; GitHub; Postman
Operating Systems	Windows 10/11; Linux; macOS

Component	Technology / Tool
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7 Software Requirements

7.1 Frontend

- React.js
- HTML
- CSS
- JavaScript

7.2 Backend

- Node.js
- Express.js

7.3 Database

- MongoDB

7.4 Authentication

- JWT
- bcrypt

7.5 Development Tools

- Visual Studio Code
- Git
- GitHub
- Postman

7.6 Operating System

- Windows 10/11
- Linux
- macOS